



How to make Desktop Neon Signs - 3D Printable, Battery or USB powered & Dimmable!



DIY Machines

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Summary

3D Print your own customisable Desktop Neon LED Sign. Powered by USB or AA batteries these will light up your room.



17.95 hrs



7 pcs



0.20 mm



0.40 mm



PLA



197 g



Prusa
MK3/S/S+

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Tags: [sign](#) [decor](#) [3dprinting](#) [lights](#) [battery](#) [lighting](#)
[usb](#) [pcb](#) [neonsign](#) [neon](#) [ledneon](#)

Please take a moment to **like** this project and **download** it from Printables - it really helps me out! Thank you

Today we're making some of these awesome customisable desktop neon signs. These ones can be powered by either a USB cable or 4 x 'AA' batteries giving you a lot of flexibility for placement.

Why not use one to illuminate your next garden or beachside party? They are also dimmable to help create the right level of ambience for any setting. Use it to help illuminate a room or dial it down for a more subtle addition to your decor.

As for customising? You can print the parts in any rigid filament colour, It's your choice as well for the colour of Neon effect LED's used.

There are several ready designed Neon shapes you can choose from and I'll be creating some more which are available on my Etsy shop: <https://www.etsy.com/shop/diymachines>

And if you're comfortable you can edit the design and create something totally unique either for yourself or as a thoughtfully made gift to someone else.

List of items used in this project and where to find them:

- Neon effect LED strips: <https://geni.us/NeonLEDStrip5v>
- Resistors 1kΩ (x2): <https://geni.us/Ufa2s>
- Capacitors 0.1μF (x2): <https://geni.us/Capacitor>
- Switching Diodes IN4148 (x2): <https://geni.us/IN4148-Diode>
- Toggle Switch DPDT (x1): <https://geni.us/DPDT-Toggle-Switch>
- Female USB Socket (x1): <https://geni.us/MiniUSBsocket>
- 50k Rotary Potentiometer (x1): <https://geni.us/Potentiometers>
- 555 timer IC (x1): <https://geni.us/555Timer>
- MOSFET Transistor (IRLB8721PBF) (x1): <https://geni.us/IRLB8721PBF>
- Bolts (M3x8mm x6): <https://geni.us/NutsAndBolts>
- Electrical hook-up wire: <https://geni.us/22AWGWire>
- Filament: <https://geni.us/3DJake> (I used PLA)
- Projects PCB: <https://geni.us/DesktopNeonSigns>

These are some of my favourite tools I use and can recommend:

- Battery-powered glue gun: <http://geni.us/BoschBatteryGlueGun>
- Bosch Bit Driver: <https://geni.us/Bosch-Screwdriver>

I use a Prusa MK3S for my projects and highly recommend both it and the Prusa Mini. If I inspired you to take up 3D printing please consider purchasing a printer via this link: <http://shop.prusa3d.com/>

#a_aid=diymachines

Prusa support my channel when you use that link at no cost to yourself. :)

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Wiring diagrams, FAQ etc can be found here: <https://www.diymachines.co.uk/3d-printed-desktop-neon-signs>

Etsy shop for 3D printable kits and more: <https://www.etsy.com/shop/diymachines>

Discord server for this project: <https://discord.gg/VjNdvW9RV6>

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SAY THANKS:

Provide continual and dependable support through Patreon: <https://www.patreon.com/diymachines>

Or join this channel on Youtube Memberships: <https://www.youtube.com/channel/UC3jc4X-kEq-dEDYhQ8QoYnQ/join>

Buy me a coffee to say thanks: <https://ko-fi.com/diymachines>

SUBSCRIBE: ■ https://www.youtube.com/channel/UC3jc4X-kEq-dEDYhQ8QoYnQ?sub_confirmation=1

INSTAGRAM: https://www.instagram.com/diy_machines

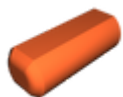
FACEBOOK: <https://www.facebook.com/diymachines/> ETSY: <https://www.etsy.com/uk/shop/DIYMachines>

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Disclaimer: This video is shared for demonstration purposes only.

Model files

desktop-neon-shape-planet.3mf



desktop-neon-toggle.stl

desktop-neon-shape-pineapple.3mf

desktop-neon-toggle.3mf



desktop-neon-shape-planet.stl

desktop-neon-housing-bottom.3mf

desktop-neon-housing-top.3mf



desktop-neon-shape-pineapple.stl



desktop-neon-housing-top.stl



desktop-neon-housing-bottom.stl

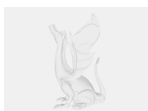
desktop-neon-dial.3mf



desktop-neon-dial.stl

desktop-neon-shape-cactus.3mf

desktop-neon-shape-cactus.stl



neon-shape-starter.stp

☐ Useful for designing your own shapes

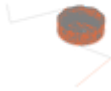
Print files

desktop-neon-toggle_02mm_pla_mk3s_6m.gcode



🌀 PLA 📏 0.40 mm ⚖️ 0.20 mm ⌚ 0.10 hrs ⚖️ 1 g 🖨️ Prusa MK3/S/S+

desktop-neon-housing-bottom.gcode



🌀 PLA 📏 0.40 mm ⚖️ 0.20 mm ⌚ 4.46 hrs ⚖️ 50 g 🖨️ Prusa MK3/S/S+

desktop-neon-housing-top_02mm_pla_mk3s_2h44m.gcode



🌀 PLA 📏 0.40 mm ⚖️ 0.20 mm ⌚ 2.74 hrs ⚖️ 33 g 🖨️ Prusa MK3/S/S+

desktop-neon-shape-pineapple_02mm_pla_mk3s_3h13m.gcode



🌀 PLA 📏 0.40 mm ⚖️ 0.20 mm ⌚ 3.21 hrs ⚖️ 34 g 🖨️ Prusa MK3/S/S+

desktop-neon-shape-planet_02mm_pla_mk3s_3h34m.gcode



🌀 PLA 📏 0.40 mm ⚖️ 0.20 mm ⌚ 3.57 hrs ⚖️ 38 g 🖨️ Prusa MK3/S/S+

desktop-neon-dial_02mm_pla_mk3s_9m.gcode



🌀 PLA 📏 0.40 mm ⚖️ 0.20 mm ⌚ 0.14 hrs ⚖️ 1 g 🖨️ Prusa MK3/S/S+

desktop-neon-shape-cactus_02mm_pla_mk3s_3h39m.gcode

🌀 PLA 📏 0.40 mm ⚖️ 0.20 mm ⌚ 3.73 hrs ⚖️ 40 g 🖨️ Prusa MK3/S/S+

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